IPL PERFORMANCE ANALYTICS

Mini Project Report

Student Name: _____

Course / Year: B.Tech – III Year

Institute: _____

Academic Year: 2026–27

1. Project Objective

The objective of this mini project is to analyse IPL match and player-performance data and convert raw records into useful sports and business insights. The analysis covers match outcomes, team performance, player contribution, toss decisions, venues, scoring patterns and season-wise trends.

2. Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Power BI Desktop, Power Query and DAX.

3. Dataset Information

The project dataset contains 1,095 match-level records across seasons 2008–2024. It follows the column structure requested in the institute assignment.

Dataset note: The included CSV is a practice dataset generated to match the assignment schema. Replace it with the official institute dataset, if supplied, before final submission.

4. Data Loading & Cleaning

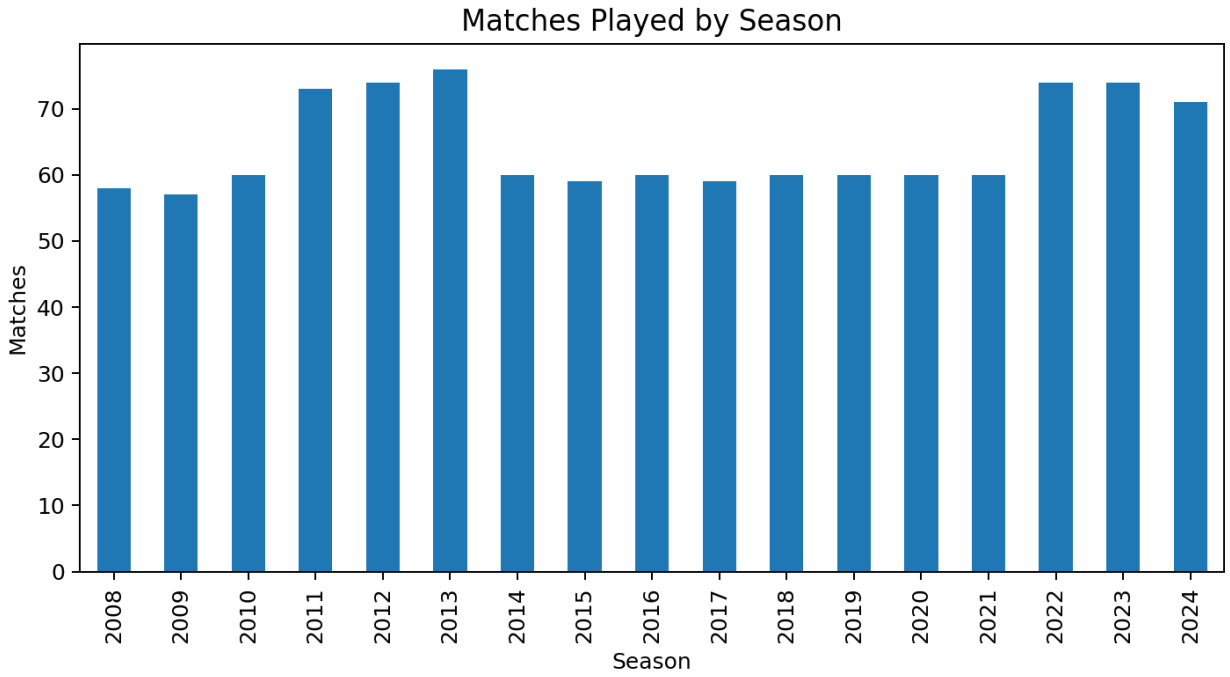
The data was loaded using Pandas. Records, columns, data types, missing values and duplicate rows were checked. Match dates were converted to datetime format and key categorical fields were reviewed for unique values.

Check	Result
Rows	1,095
Columns	26
Duplicate rows	0
Missing values	0
Date field	Datetime

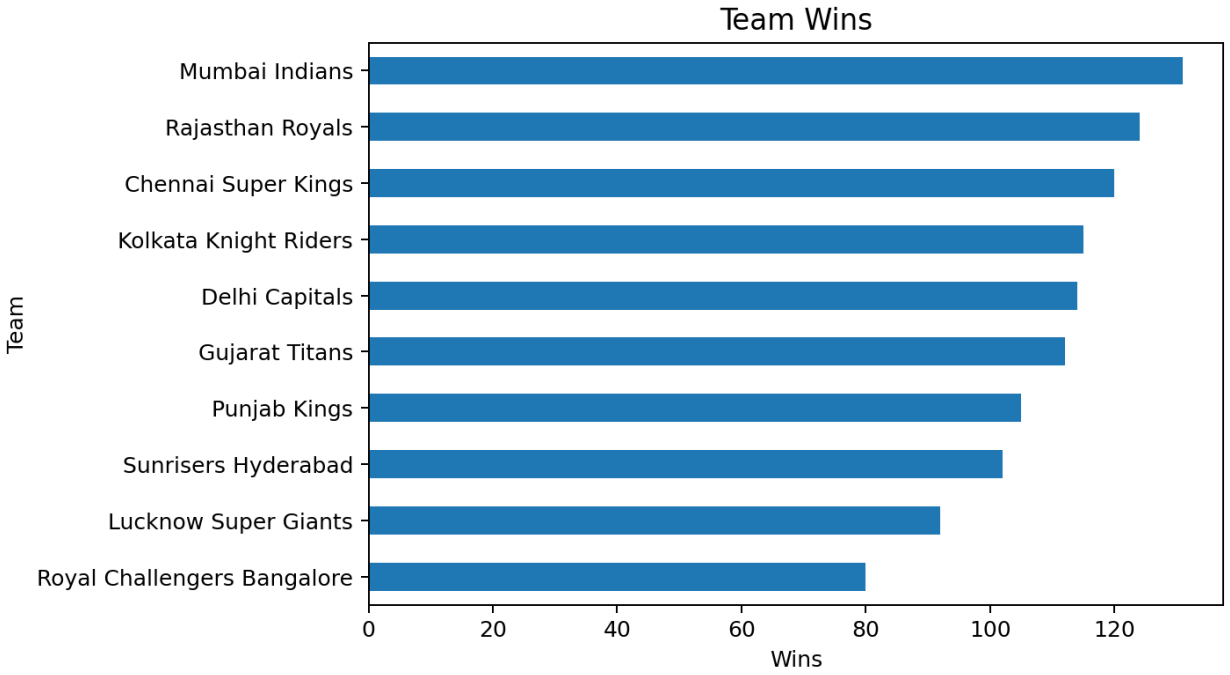
5. Key Analysis Results

Metric	Result
Total matches	1,095
Highest-win team	Mumbai Indians
Most POM awards	Abhishek Sharma
Most frequent venue	MA Chidambaram Stadium
Highest featured score	Ruturaj Gaikwad – 109
Highest run margin	60
Highest wicket margin	10
Toss-winner match-win rate	50.05%

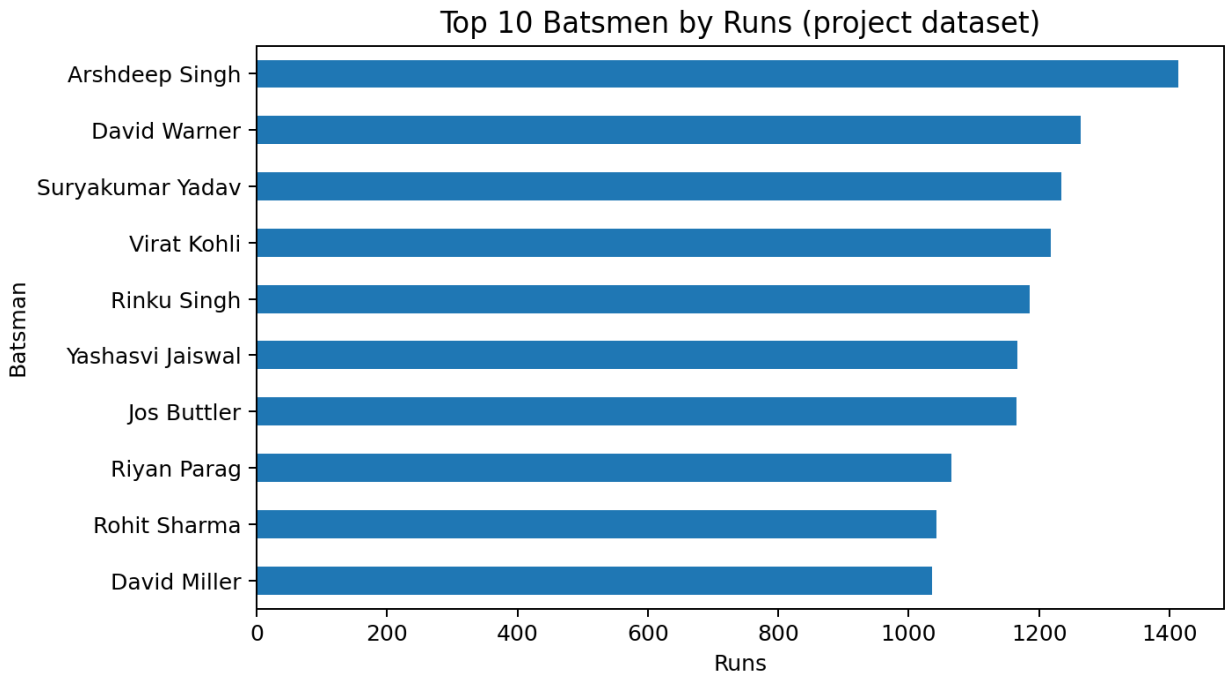
6. Matches by Season



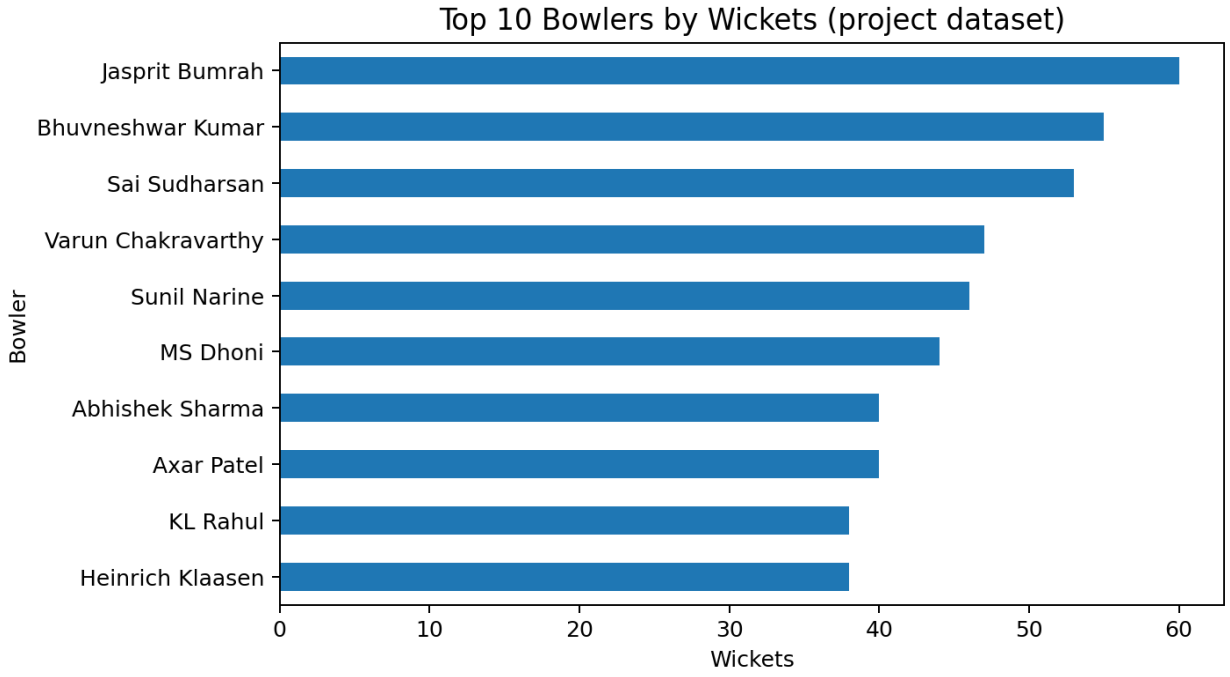
7. Team Wins



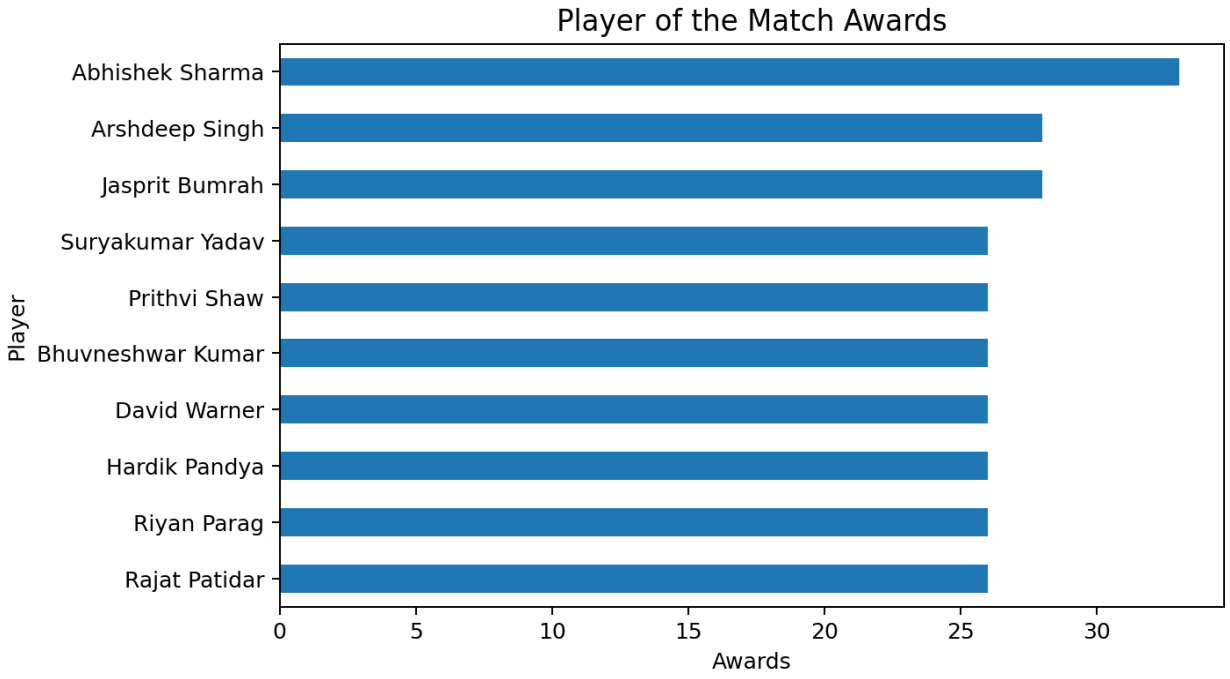
8. Top Run Scorers



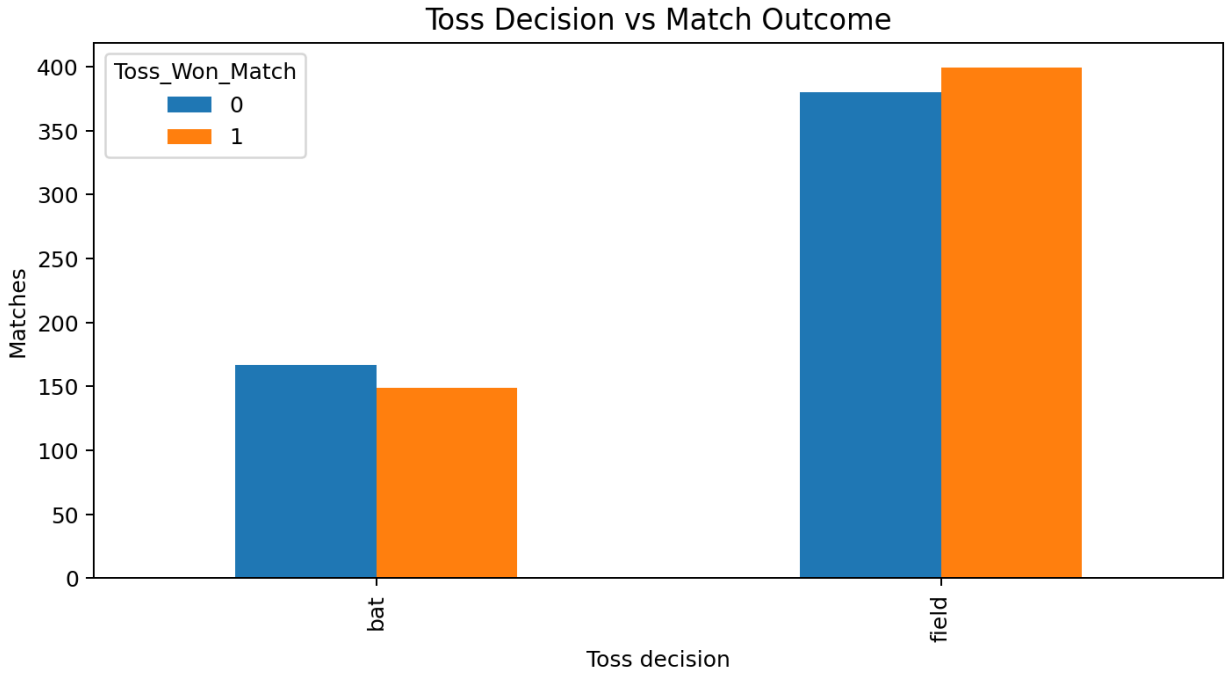
9. Top Wicket Takers



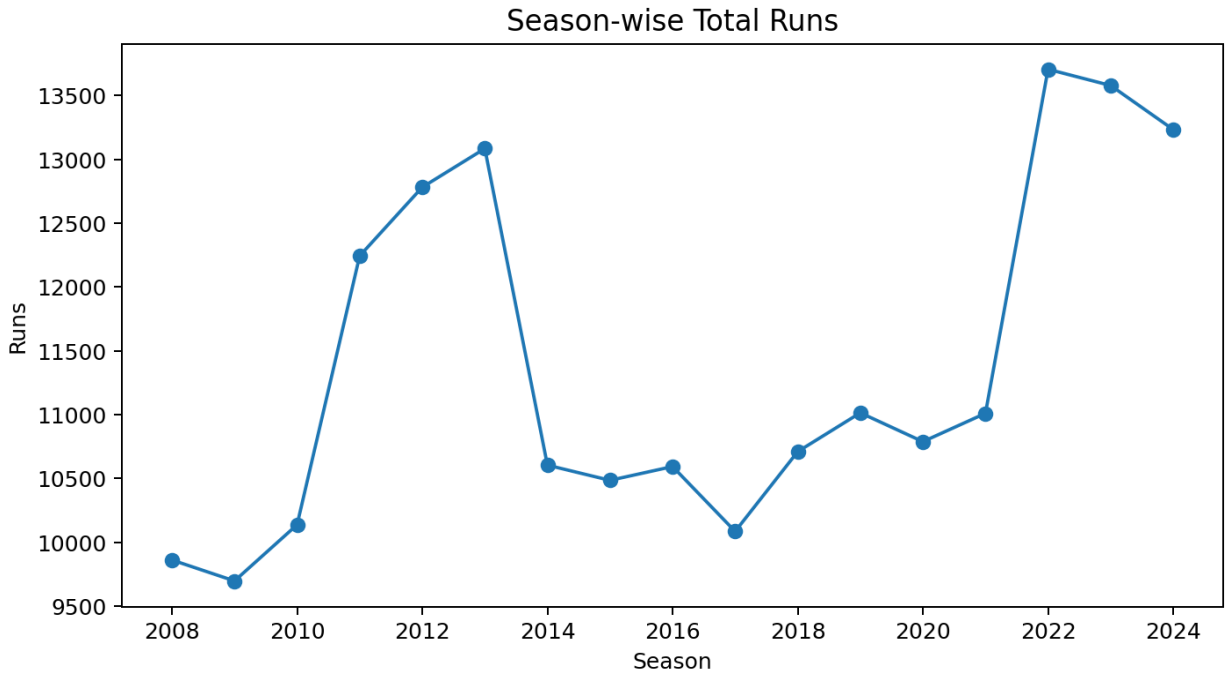
10. Player of the Match Awards



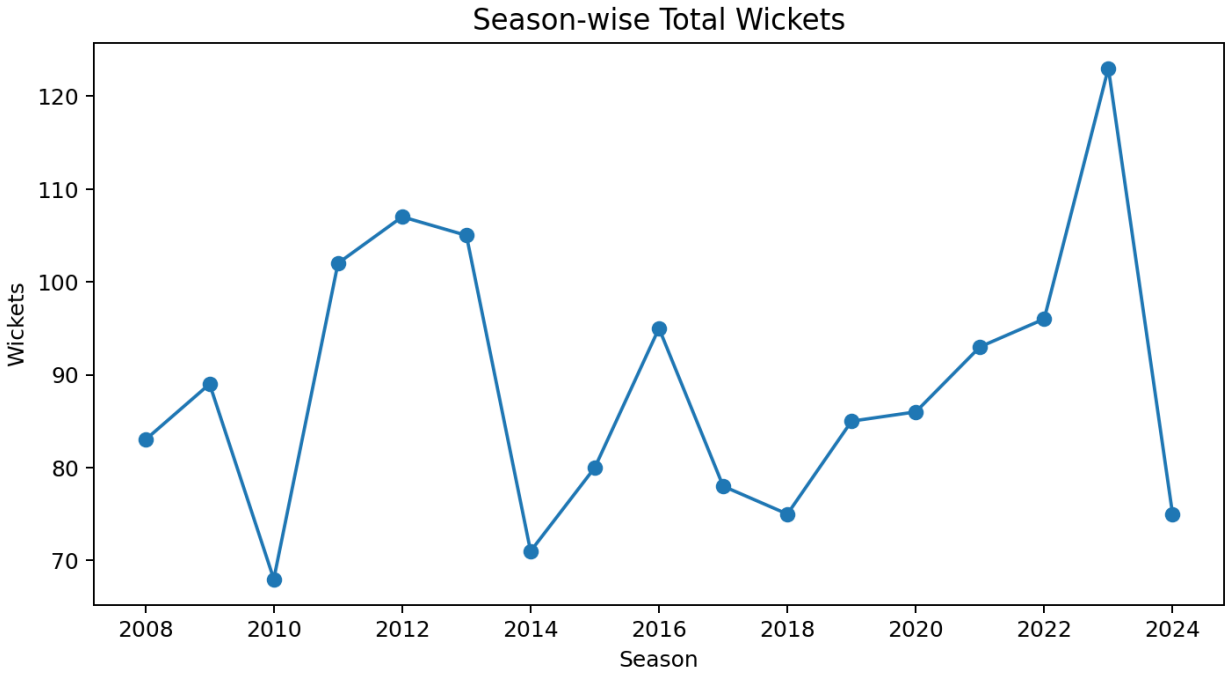
11. Toss Decision Analysis



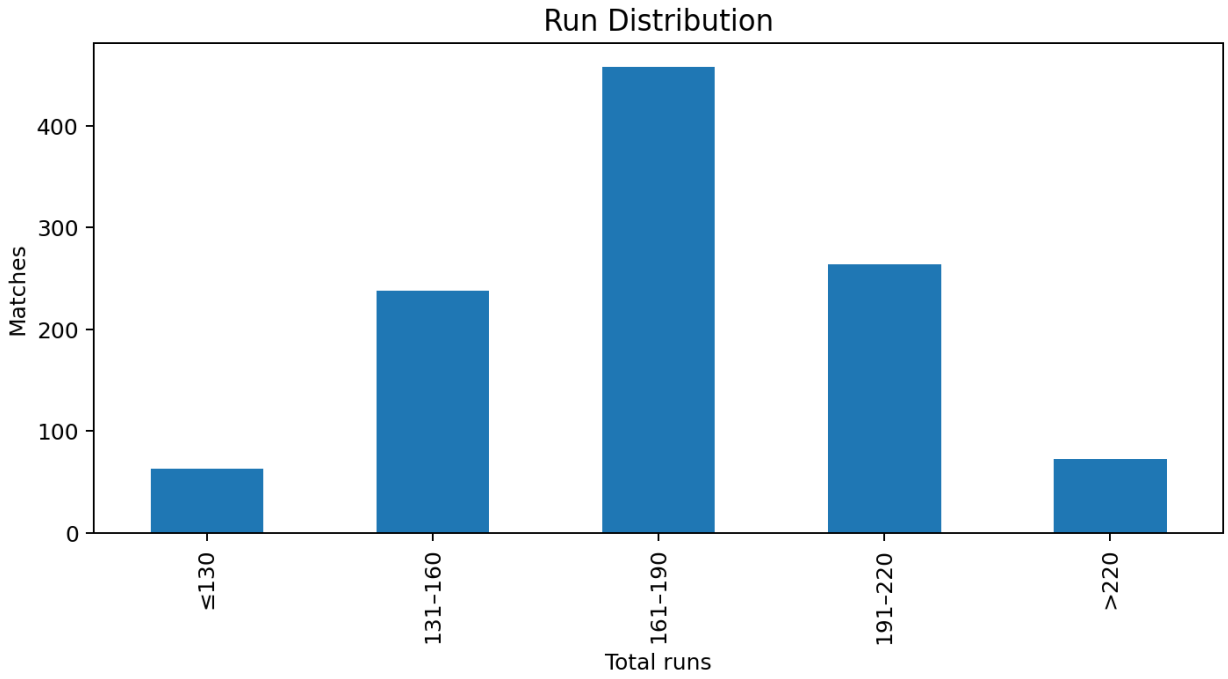
12. Season-wise Total Runs



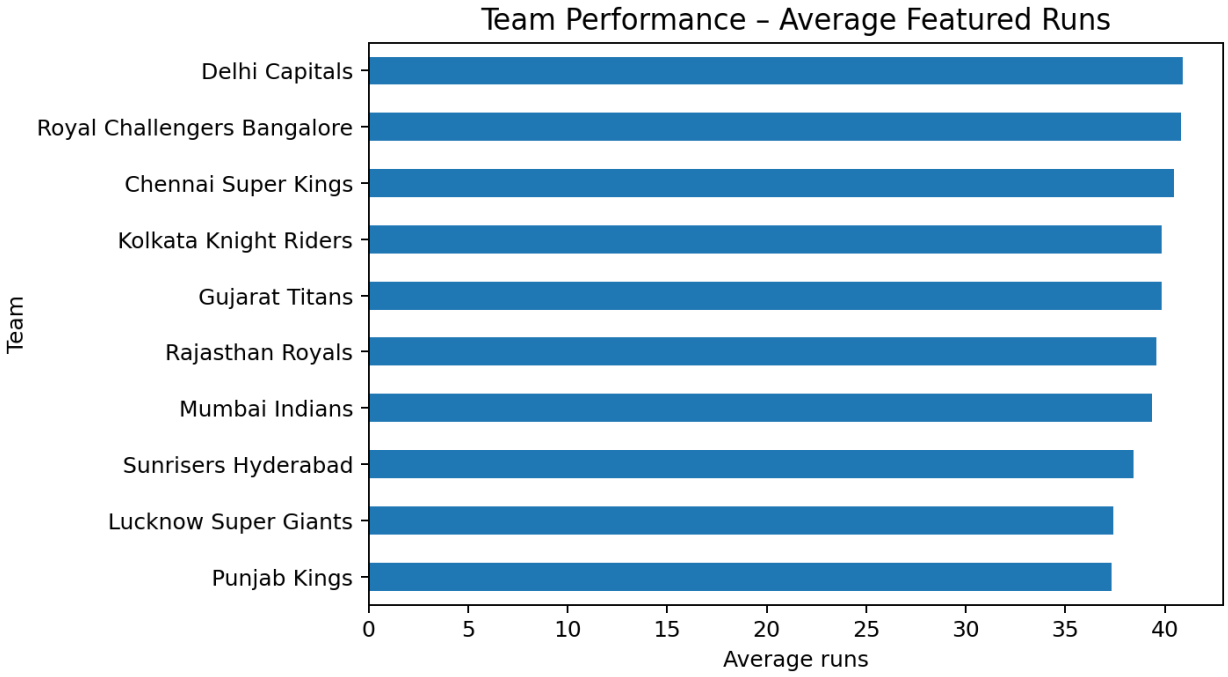
13. Season-wise Total Wickets



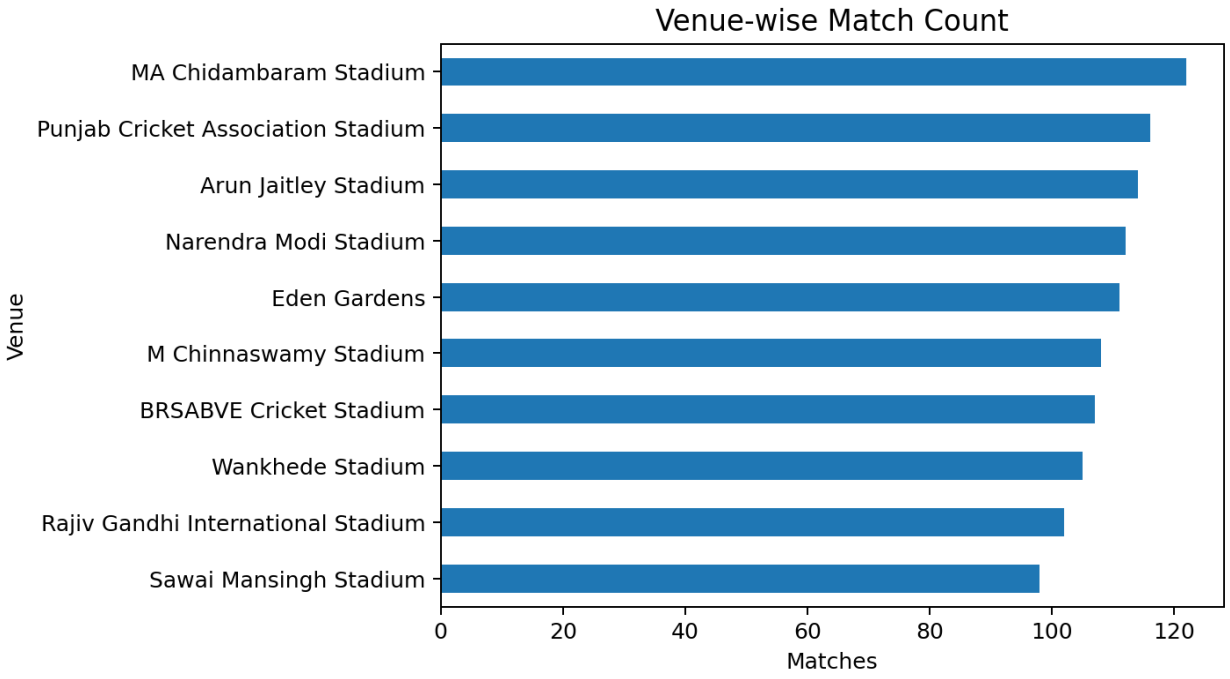
14. Run Distribution



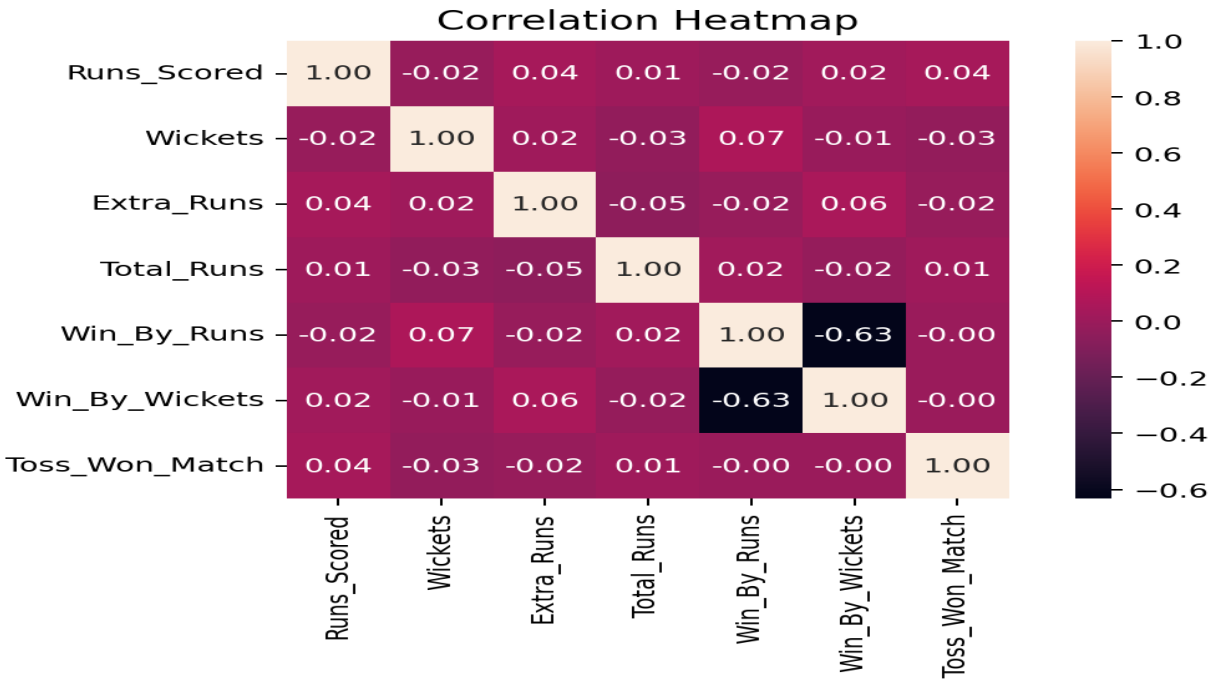
15. Team Performance



16. Venue-wise Match Count

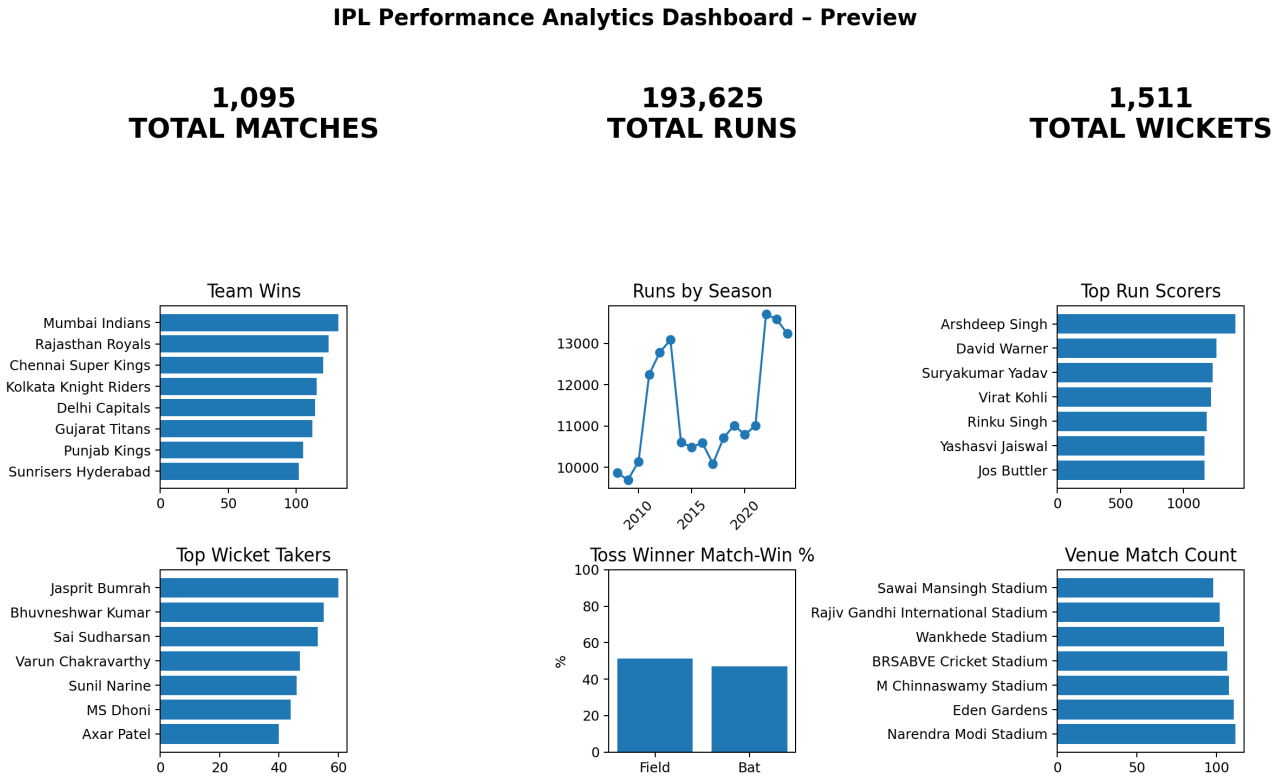


17. Correlation Heatmap



18. Power BI Dashboard

The dashboard layout contains KPI cards, team and player charts, season trends, venue analysis and slicers for Season and Venue.



19. Business Insights

• The dataset covers 1,095 match records across 17 seasons.
• Mumbai Indians has the highest number of wins in the project dataset.
• Abhishek Sharma has the highest Player-of-the-Match count.
• MA Chidambaram Stadium is the most represented venue.
• The toss-winner match-win rate is 50.05%.
• Season-wise totals show variation in scoring volume across seasons.
• Player grouping highlights differences in batting and bowling contribution.

20. Conclusion

The project demonstrates an end-to-end sports analytics workflow using Python and prepares the same data for interactive Power BI reporting. The analysis can be extended with the official institute dataset for final submission.

21. Submission Checklist

■ Python notebook ■ Cleaned CSV ■ Power BI dashboard ■ Project report ■ README ■ DAX measures

Student Signature: _____ Date: _____